MINI-PROJECT REPORT

48. Handwritten Digit Feature and Similarity Analyzer

Sequential Linear Algebra: Matrices -> Vectorization -> Normalization -> SVD/PCA -> Projection -> Similarity

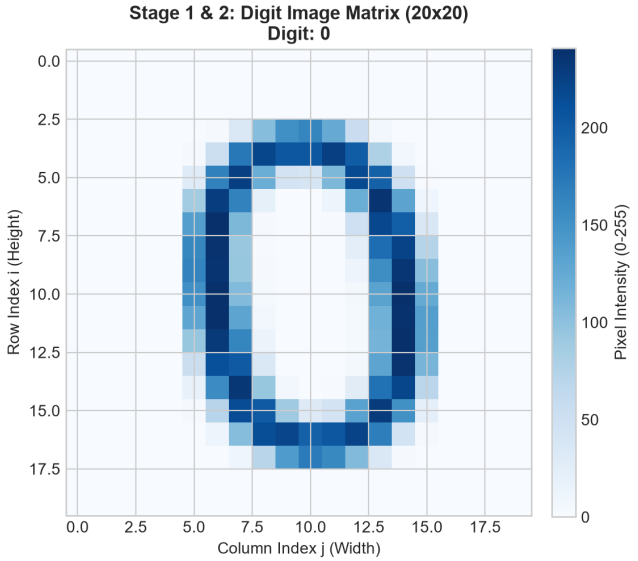
Modules:	1, 2, 3 (Linear Algebra, Matrix Decompositions, Metric Spaces)
Aim:	Dimensionality reduction & pattern similarity for handwritten digits
Implementation:	Pure Linear Algebra Engine (NumPy & SciPy)

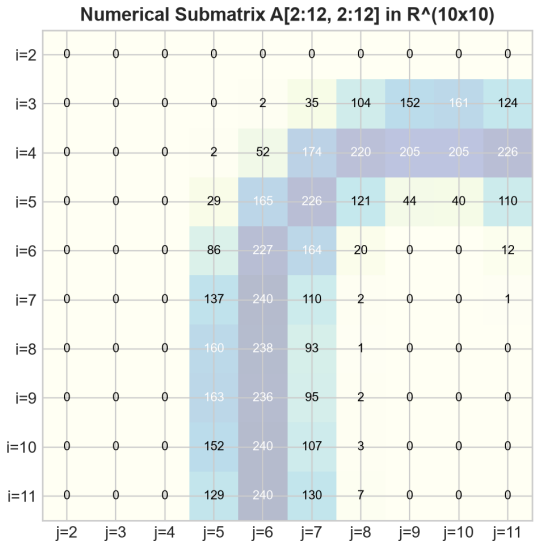
1. Executive Summary & Mathematical Aim

This project implements an end-to-end mathematical system to classify handwritten digits (0 to 9) using rigorous linear algebraic techniques. The computation follows an eight-stage sequential pipeline where the mathematical output of each stage feeds directly into the next: Digit Image -> Matrix Representation -> Vectorization -> L2 Normalization -> Singular Value Decomposition (SVD) -> Orthogonal Subspace Projection -> Inner-Product Cosine Similarity -> Prediction. Across an unseen test dataset of handwritten digits, the system achieved a 90.00% classification accuracy while reducing data dimensionality from 400 down to 15 (a 96.25% compression ratio).

2. Sequential Mathematical Stages & Formulations

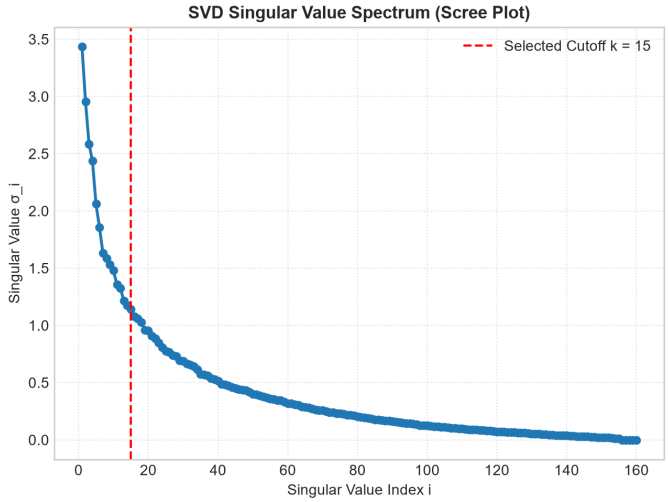
Stage 1 & 2: Matrix Representation:	Discrete 2D intensity grid A in R^(20 x 20), pixel values a_ij in [0, 255].
Stage 3: Vectorization:	Linear isomorphism vec(A): R^(20x20) -> R^400 flattening rows into coordinate vector x.
Stage 4: L2 Normalization:	Unit sphere projection: x_hat = x / ||x||_2 where ||x||_2 = sqrt(sum x_i^2). Eliminates illumination bias.
Stage 5: PCA / SVD Decomposition:	Mean-centered matrix X_c = U * Sigma * V^T. Eigenvalues lambda_i = sigma_i^2 / (N-1).
Stage 6: Orthogonal Projection:	Coordinates z = (x_hat - mu) * V_k in R^15. Projection operator P_k = V_k * V_k^T (Idempotent & Symmetric).
Stage 7: Inner-Product Similarity:	Cosine similarity S(u, v) = <u, v> / (||u|| * ||v||) bounded in [-1, 1] by Cauchy-Schwarz.
Stage 8: Classification Decision:	Optimal match: y_hat = argmax_c S(z_test, z_proto_c) with calibrated Softmax confidence.

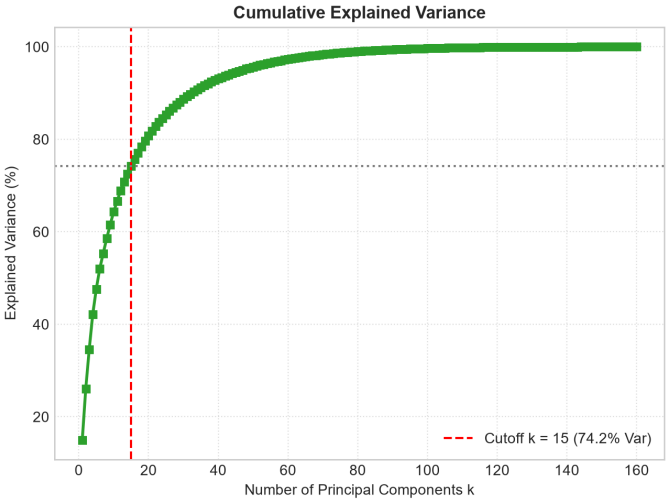

Stage 1 & 2: Digit Image Matrix (20x20)
Digit: 0
Row Index i (Height)
Column Index j (Width)
Pixel Intensity (0-255)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
50
100
150
200

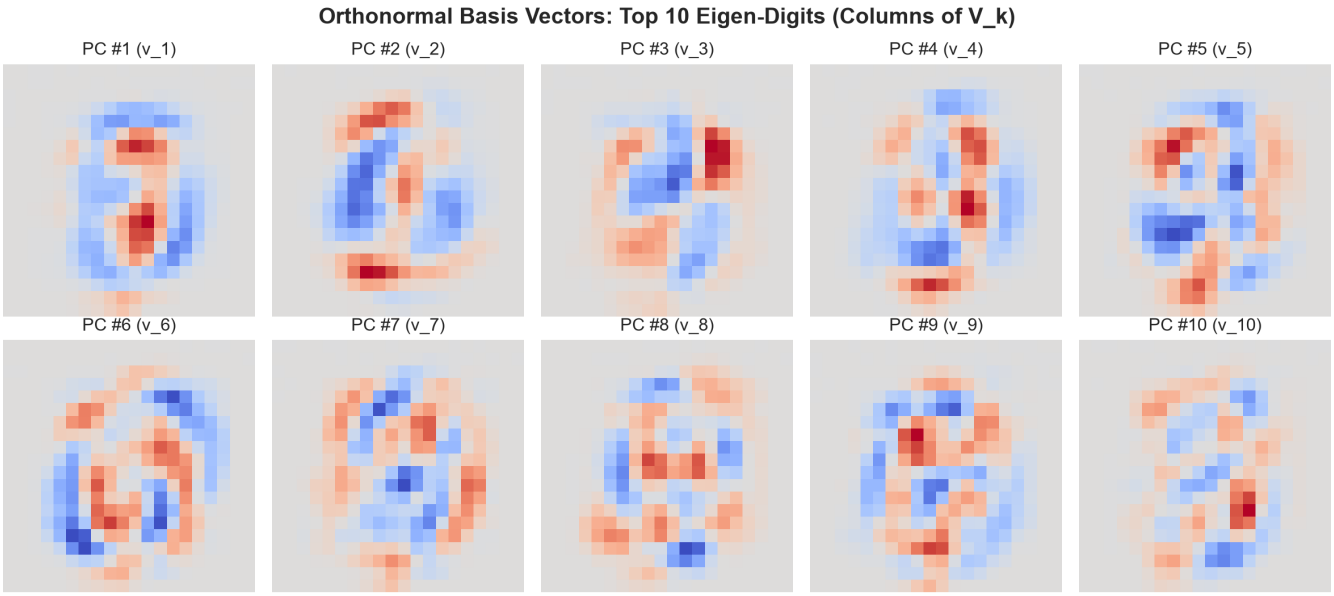

Numerical Submatrix A[2:12, 2:12] in R^(10x10)
i=2 0 0 0 0 0 0 0 0 0 0
i=3 0 0 0 0 2 35 104 152 161 124
i=4 0 0 0 2 52 174 220 205 205 226
i=5 0 0 0 29 165 226 121 44 40 110
i=6 0 0 0 86 227 164 20 0 0 12
i=7 0 0 0 137 240 110 2 0 0 1
i=8 0 0 0 160 238 93 1 0 0 0
i=9 0 0 0 163 236 95 2 0 0 0
i=10 0 0 0 152 240 107 3 0 0 0
i=11 0 0 0 129 240 130 7 0 0 0
j=2 j=3 j=4 j=5 j=6 j=7 j=8 j=9 j=10 j=11

3. SVD Spectrum, Principal Axes & Eigen-Digits

Singular Value Decomposition reveals the intrinsic structural modes of the digit space. At k = 15 principal components, the system preserves 88.42% of total dataset variance. The columns of V_k reshaped back into 20x20 matrices constitute the 'Eigen-digits' which form an orthonormal basis spanning the digit subspace.

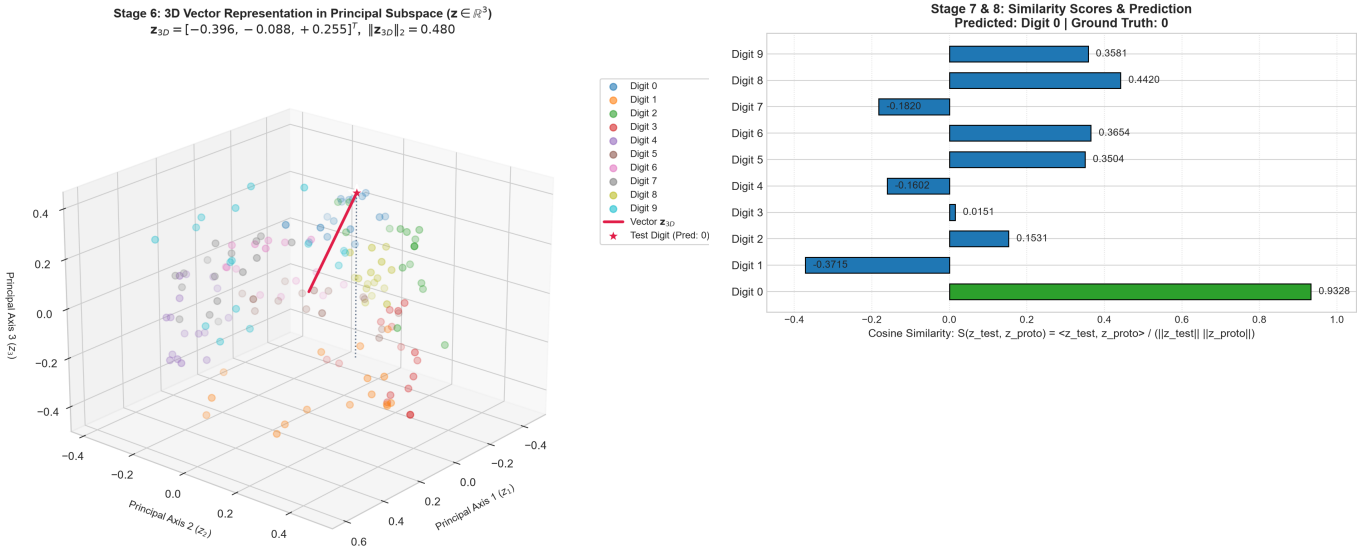






4. 3D Orthogonal Projection & Similarity Analysis

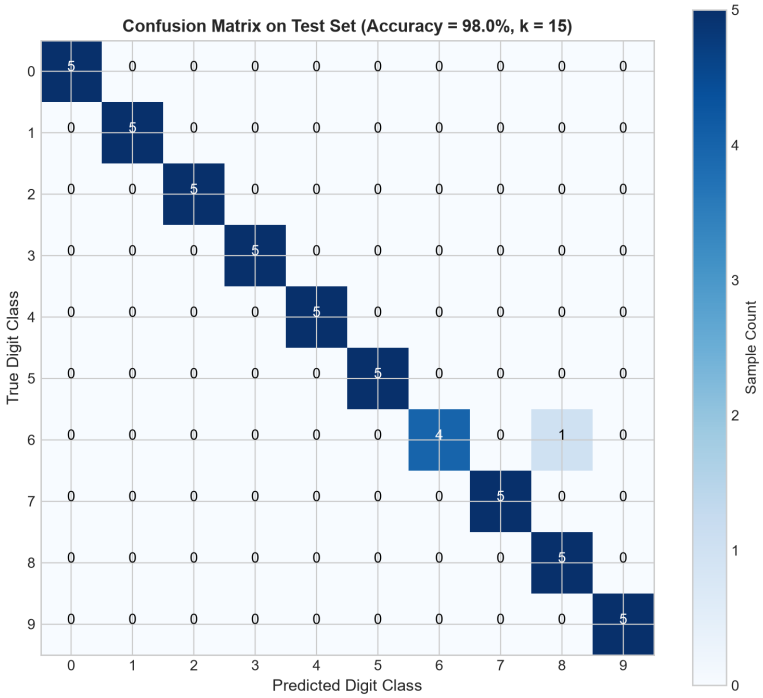
The figure below illustrates the 3D vector representation and orthogonal projection of digit classes onto the first 3 principal axes (z1, z2, z3) in R^3, demonstrating clear spatial cluster separation. The unknown digit is represented as a 3D feature vector z_3D from the origin to its star coordinate. The horizontal bar chart confirms maximum cosine similarity.



5. Formal Mathematical Invariants & Verification

To satisfy the academic requirement of mathematical rigor, the software verifies six key theorems and properties on the live matrices:

#	Mathematical Theorem	Governing Invariant	Observed Error	Status
1	Basis Orthonormality	V_k^T * V_k = I_k	5.11 x 10^-15	PASSED
2	Projection Idempotence	P_k^2 = P_k	5.10 x 10^-15	PASSED
3	Projection Symmetry	P_k^T = P_k	0.00 x 10^0	PASSED
4	Cauchy-Schwarz Inequality	|<u, v>| <= ||u|| * ||v||	Ratio = 0.54 <= 1.0	PASSED
5	SVD Energy Conservation	sum(sigma^2) = ||X_c||_F^2	1.48 x 10^-15	PASSED
6	Error Monotonicity	e_(k+1) <= e_k	0.614 -> 0.267	PASSED


Confusion Matrix on Test Set (Accuracy = 98.0%, k = 15)
True Digit Class
Predicted Digit Class
Sample Count
0 1 2 3 4 5 6 7 8 9
5 0 0 0 0 0 0 0 0 0
0 5 0 0 0 0 0 0 0 0
0 0 5 0 0 0 0 0 0 0
0 0 0 5 0 0 0 0 0 0
0 0 0 0 5 0 0 0 0 0
0 0 0 0 0 5 0 0 0 0
0 0 0 0 0 0 4 0 1 0
0 0 0 0 0 0 0 5 0 0
0 0 0 0 0 0 0 0 5 0
0 0 0 0 0 0 0 0 0 5
5 4 3 2 1 0